

The Emergency Access Health Data System (EA-HDS)

Introduction

The Emergency Access Health Data System (EA-HDS) is developed to offer secure, rapid, and controlled access to life-saving medical data during emergencies. It is designed to give certified first responders access to essential health details like allergies, medications, or chronic illnesses without compromising the complete Electronic Health Records (EHRs). The system emphasizes selective disclosure, real-time response, strong security, auditability, and patient-driven control.

System Components

- Encrypted Health Data Vault (EHDV): Stores encrypted patient health records securely in the cloud.
- Emergency Access Interface (EAI): A secure API that handles access requests from certified responders.
- Patient Policy Dashboard (PPD): A web/mobile platform enabling patients to define and manage emergency access rights.
- Distributed Ledger System (DLS): A blockchain network used to store immutable access logs and access policies.
- Responder Access Devices (RAD): Secure devices used by responders, equipped with biometric authentication and a Trusted Execution Environment (TEE).
- Secure Hardware Module (SHM): Embedded in responder devices to securely decrypt and display sensitive data.

Protocol Workflow

Step 1: Patient Policy Definition

- Patients use the PPD to specify what data can be accessed in emergencies.
- Policies are hashed and stored in the DLS.
- The EHR is encrypted with AES-256 and segmented into emergency-accessible subsets.
- Emergency decryption keys are generated and protected via threshold encryption (e.g., 2-of-3 authority model).

Step 2: Emergency Access Request

- Responder authenticates with biometrics and a valid certificate on the RAD.
- Patient ID is captured (via QR, NFC, or biometric match) and sent to the EAI.
- The TEE verifies the request context before submission.

Step 3: Policy Validation and Key Retrieval

- EAI checks DLS for policy match.
- If policy conditions are satisfied, threshold decryption is triggered by trusted authorities.
- The decrypted emergency key is used in the TEE to access only approved data.

Step 4: Access Logging and Auditing

- Every access is logged (who, when, where, what) and signed digitally.
- Logs are appended to the DLS and are visible to patients via the PPD.
- Unauthorized access can be flagged and revoked in real-time.

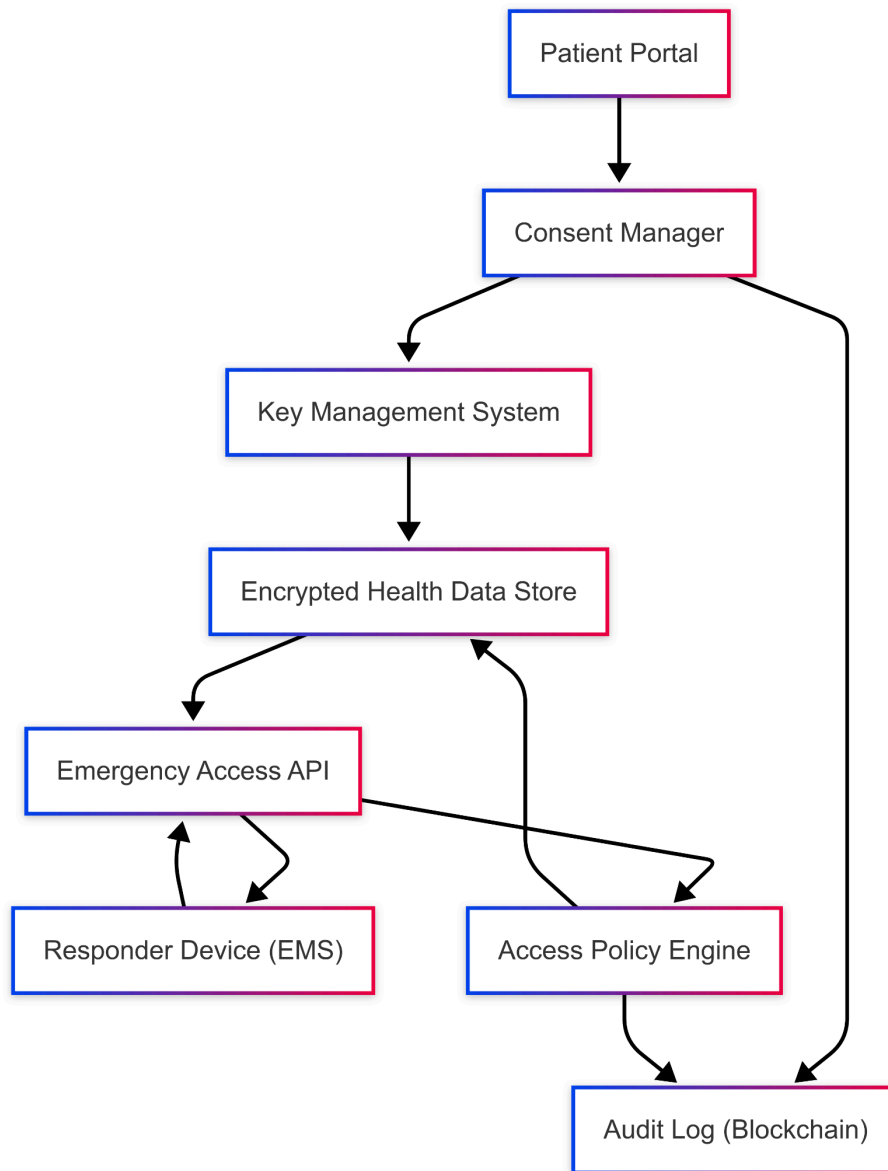
Step 5: Patient-Centric Control

- Patients can update policies, disable access, or revoke responder rights via the PPD.
- Wearables with QR/NFC link responders to the latest policy versions

Security Design

Requirement	Feature Used	Purpose
Selective Disclosure	Data segmentation + threshold decryption	Ensures only emergency-specific data is decrypted.
Rapid Access	Local TEE + AES-256	Enables quick decryption within responder devices.
Security Assurance	AES-256, TEE, blockchain logging	Protects EHR integrity, logs unauthorized access attempts.
Accountability	Blockchain + digital signatures	Provides tamper-proof access audit logs.
Patient Control	PPD + wearable-linked policies	Grants patients continuous control over data access permissions.

System Architecture



Trade-Offs and Considerations

1. **Offline Cache:**
 - RADs may store encrypted emergency data caches for offline use.
2. **Latency vs. Security:**
 - ensures fast decryption; threshold retrieval adds brief delay.
3. **Scalability:**
 - Decentralized DLS allows scaling across institutions.
4. **Privacy Assurance:**
 - TEE and data scoping prevent data leakage or duplication.

Conclusion

This conceptual architecture satisfies all the specified requirements for an emergency health data access system. It employs advanced cryptography, secure hardware, decentralized trust, and patient-oriented policy control to ensure that critical data is available instantly during emergencies without compromising long-term privacy and security.